KRRISH BHASIN

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EDUCATION

BITS Pilani, K.K. Birla Goa Campus — B.E. (Hons.) Computer Science *Expected May 2028*

Coursework: Linear Algebra, Probability & Statistics, Foundations of Data Science, Machine Learning, OOP *Goa, India*

- Codeforces Specialist (peak **1550**, 790+ solved) · Core Member, AlgoManiax Competitive Programming Club

EXPERIENCE

ML Research Intern — Indian Institute of Technology Bombay *(NLP / MT Lab, Prof. Ganesh Ramakrishnan)* *Sep 2025 – Feb 2026*

- Improved a LLaMA-based LLM post-correction pipeline with domain-specific glossary injection, raising BLEU from **0.211 to 0.271 (+28%)** on internal benchmarks for technical and rare-domain vocabulary.
- Fine-tuned IndicTrans2 (Fairseq) on 1.2M sentence pairs across 12 Indic language pairs; ran fine-tuned vs zero-shot ablations (BLEU, chrF) and presented results to the lab.
- Diagnosed and fixed tokenization mismatches destabilizing training in the Udaan OCR/MT pipeline for digitizing Indian-language educational content.

Software Engineering Intern — Hodo Stays *May 2026 – Jul 2026*

LLM-powered guest communication for a hospitality SaaS platform *Bengaluru, India*

- Shipped a multi-channel (chat, email, in-app) guest messaging system that automates routine communication, using prompt templates with dynamic context injection (guest name, property, booking status) for brand-aligned, grounded responses.
- Engineered a three-class intent classifier (booking / maintenance / general) reaching **~91%** accuracy, cutting manual message triage.
- Built the backend on Supabase (PostgreSQL, real-time subscriptions) and routed model calls through OpenRouter, maintaining **~1.4s** median response time.

PROJECTS

Mixture-of-Experts Layer for Named Entity Recognition | *Python, PyTorch, HuggingFace* — Github

- Built a Mixture-of-Experts layer from scratch (4 expert MLPs, top-2 gating, load-balancing loss) and added it to a BiLSTM tagger with GloVe, character-CNN and casing features.
- Tested on CoNLL-2003 over 3 seeds against a plain BiLSTM and a parameter-matched dense-FFN control. The balanced MoE matched the baseline (**89.3 ± 0.4 vs 89.4 ± 0.3** entity-F1) and beat the unbalanced MoE by about 0.9 F1, with expert usage spread evenly (17–33% per expert).

AlphaZero for Hex | *Python, JAX, Flax* — *Github*

- Implemented AlphaZero in JAX with a JIT-compiled, vmap-batched MCTS running 256 self-play games in parallel, trained on a MacBook Air CPU.
- Matched MoHex 2.0 (Computer Olympiad champion) on all 25 winnable 5x5 openings and **99.2%** (768/774) of 7x7 positions, verified with MoHex's proof-number solver; reproduced Jones (2021)'s train- vs test-time compute trade-off (2.5x less training).

Speculative Decoding for LLM Inference | *Python, PyTorch, HuggingFace* — *Github*

- Implemented speculative decoding (Leviathan et al., 2023) from scratch, including custom KV-cache management for draft/target models, and validated correctness via accept/reject sanity checks.
- Benchmarked on CPU and GPU (T4) across 20 prompts against autoregressive decoding; identified model scale as the key driver of hardware-dependent speedup.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, Bash

ML / AI: PyTorch, TensorFlow, HuggingFace Transformers, Fairseq, LoRA / PEFT, Scikit-learn, NumPy

LLM & NLP: RAG pipelines, FAISS, prompt engineering, IndicTrans2, BLEU / chrF / ROUGE evaluation